# MODULE B
# Reservoir Storage Estimation

*Complete Research Planning and Technical Discussion Brief*

Primary focus: Satellite-based estimation of changes in reservoir water storage | Secondary focus: exploratory absolute-storage estimation

Prepared for supervisor, team and irrigation-engineer discussions | Status: pre-validation; data collection and methodological design in progress

**How to read this document** Verified published evidence, teammate-reported information and proposed research choices are intentionally distinguished. No computational model has yet been validated.

## Executive decision summary

- Main question: Can SWOT-derived water surface elevations and Module A-derived water surface areas produce defensible estimates of storage change in Sri Lankan reservoirs, with quantified uncertainty?
- Baseline methods: trapezoidal integration and PCHIP-based area–elevation integration. Proposed advanced experiment: physically constrained, uncertainty-aware modelling; no method is selected as best in advance.
- Validation design: use independent major-reservoir storage records; then examine transferability to small tanks, provided suitably matched small-tank reference measurements are obtained.
- Absolute volume is exploratory: satellite heights and areas identify changes between observed levels, not a uniquely known total volume without a storage anchor, capacity relationship or bathymetric assumptions.
- IT deliverable: a reproducible Python pipeline, comparative experiments, uncertainty quantification, validation outputs and optional API/dashboard. Machine learning is optional, not a prerequisite.

## 1. Purpose, problem and research gap

### 1.1 Purpose

Develop and evaluate a computational framework that converts dated reservoir water-surface elevations and surface areas into estimates of change in stored water volume. The framework should be transparent about uncertain observations and tested against independent reference records. Where supporting reference information exists, explore estimation of absolute water storage and percentage of capacity.

### 1.2 Problem statement

Many small irrigation tanks lack readily usable, frequently updated area–elevation–capacity curves or continuous storage monitoring. Satellite observations can provide partial measurements, but they are temporally sparse and have quality limitations. Neither a pair of water levels nor two surface-area measurements uniquely reveals the complete underwater geometry between those levels. A defensible method must therefore model the area–elevation relationship and report uncertainty.

### 1.3 Literature-grounded gap and proposed novelty

It would be incorrect to claim that reservoir volume estimation has never been performed in Sri Lanka, or that satellite-based storage change is a new concept. The Irrigation Department publishes major-reservoir storage values, Imbulana (2022) studied village-tank depth–area–storage relationships, and SWOT has existing lake-storage-change products. The plausible contribution is the controlled comparison and evaluation of methods using combined SWOT elevations and Module A areas under sparse observations, irregular geometry and uncertainty, with a Sri Lankan major-to-small-reservoir transferability study. A comprehensive literature comparison is still required before fixing a novelty claim. [1–3]

## 2. Scope, objectives and module boundaries

| Element | Definition / responsibility |
|---|---|

| | |
|---|---|
| **Primary objective** | Estimate net reservoir storage change ΔV between two observed water elevations. |
| **Secondary objective** | Explore absolute storage V(h) only with defensible reference information or explicit assumptions and uncertainty. |
| **Module A (teammate)** | Derive water-surface areas / segmentation masks and report area quality or uncertainty; Module B does not redesign segmentation. |
| **Module B (this project)** | Retrieve/quality-screen SWOT heights; pair observations; model A(h); integrate storage change; quantify uncertainty; independently validate; build computational prototype. |
| **Out of scope by default** | Direct rainfall-runoff prediction, reconstructing full 3D bathymetry for every lake, real-time daily storage where satellite data are missing, and claiming exact absolute volume without an anchor. |

### 2.1 Research questions

1. RQ1 — How accurately can storage change be estimated from matched SWOT water elevations and satellite-derived surface areas compared with independent reservoir storage records?
2. RQ2 — How do trapezoidal, PCHIP and physically constrained methods perform when area–elevation observations are sparse, noisy or unevenly distributed?
3. RQ3 — How reliable are uncertainty intervals, and how does performance vary by height gap, temporal gap, reservoir and water-level range?
4. RQ4 — Can the validated modelling procedure be transferred to small Sri Lankan irrigation tanks when their own observations are used for fitting?
5. Exploratory RQ5 — What absolute-storage information can be inferred when reference storage, rated capacity or bathymetry is incomplete?

## 3. Data inventory and evidence status

### 3.1 Five selected major reservoirs

Published independent reference evidence: Table 6-2 of the Sri Lanka Irrigation Department Administration Report 2024 lists storage at five reporting dates in 2024 for all five selected major reservoirs. These are published reference values, not yet matched to individual SWOT overpasses. [1]

| Reservoir | 01 Jan 2024 (MCM) | 01 Aug 2024 (MCM) | 31 Dec 2024 (MCM) |
|---|---|---|---|
| **Mahakandarawa** | 47.39 | 11.72 | 35.03 |
| **Huruluwewa** | 69.07 | 19.64 | 34.72 |
| **Nachchaduwa** | 55.69 | 28.42 | 52.56 |
| **Nuwara Wewa** | 44.47 | 25.90 | 39.88 |
| **Manankattiya** | 5.96 | 1.63 | 5.90 |

MCM = million cubic metres. Example reference difference for Mahakandarawa between 01 January and 01 August 2024: 11.72 − 47.39 = −35.67 MCM. This is a reported storage difference, not a satellite prediction. The official storage definition and reference datum need to be confirmed before comparison.

### 3.2 SWOT feasibility information from the team proposal

The earlier team proposal reportedly identified 13 candidate lakes/tanks: five major reservoirs and eight village tanks around Mahakanadarawa, with historical usable SWOT observations. As the underlying proposal is not available for reinspection in this document-production session, counts and IDs below are team-reported planning inputs, not independently revalidated here.

| Reservoir | Team-reported SWOT lake ID | Usable / best-quality |
|---|---|---|
| **Nachchaduwa** | 4580051323 | 76 / 55 |

| | | |
|---|---|---|
| **Hurulu Wewa** | 4580013372 | 77 / 71 |
| **Nuwara Wewa** | 4580009523 | 62 / 53 |
| **Mahakanadarawa** | 4580006192 | 85 / 63 |
| **Manankattiya** | 4580013332 | 71 / 57 |

Village tanks in the team plan: Tank 1 (4580005172, 73 usable); Tank 2 (4580010522, 73); Tank 3 (4580006282, 73); Tank 4 (4580010542, 86); Tank 5 (4580011762, 77); Tank 6 (4580011422, 90); Tank 7 / Bulankulama (4580007392, 47); and Tank 8 (4580043692, 89). The same earlier discussion suggested that the small-tank observations were mostly quality-flagged as suspect. All counts, dates, labels and flags require an original-source check before use as measured evidence.

### 3.3 Data required and collection status

| Input | Purpose | Current status / next verification |
|---|---|---|
| **SWOT LakeSP elevations + quality fields** | Water elevation h(t), observation dates and reliability | Candidate IDs and counts reported; raw files, datum, quality flags and exact timestamps to retrieve. |
| **Module A area output** | A(t) for same reservoir and close observation time | To be supplied by teammate; masks, uncertainty and matching dates not yet verified. |
| **Irrigation Department storage histories** | Independent ΔV reference | 2024 five-date table verified; daily-series coverage and individual-date provenance require checking. |
| **Small-tank storage information** | Independent transferability test | Teammate reports two small tanks may have data; identity, dates, units and whether time-series or capacity-only are unconfirmed. |
| **Capacity curves / bathymetry / engineering records** | Absolute-storage anchor and geometric reference | Availability, reference datum and definitions to confirm with engineer. |

## 4. Visual explanation — from satellite observation to storage estimate

The approved infographic summarizes the whole research concept. Its reservoir cross-section is a conceptual illustration, not a measured shape; the example curve is illustrative, not observed data. Read the equations in Section 5 for the exact proposed calculations.

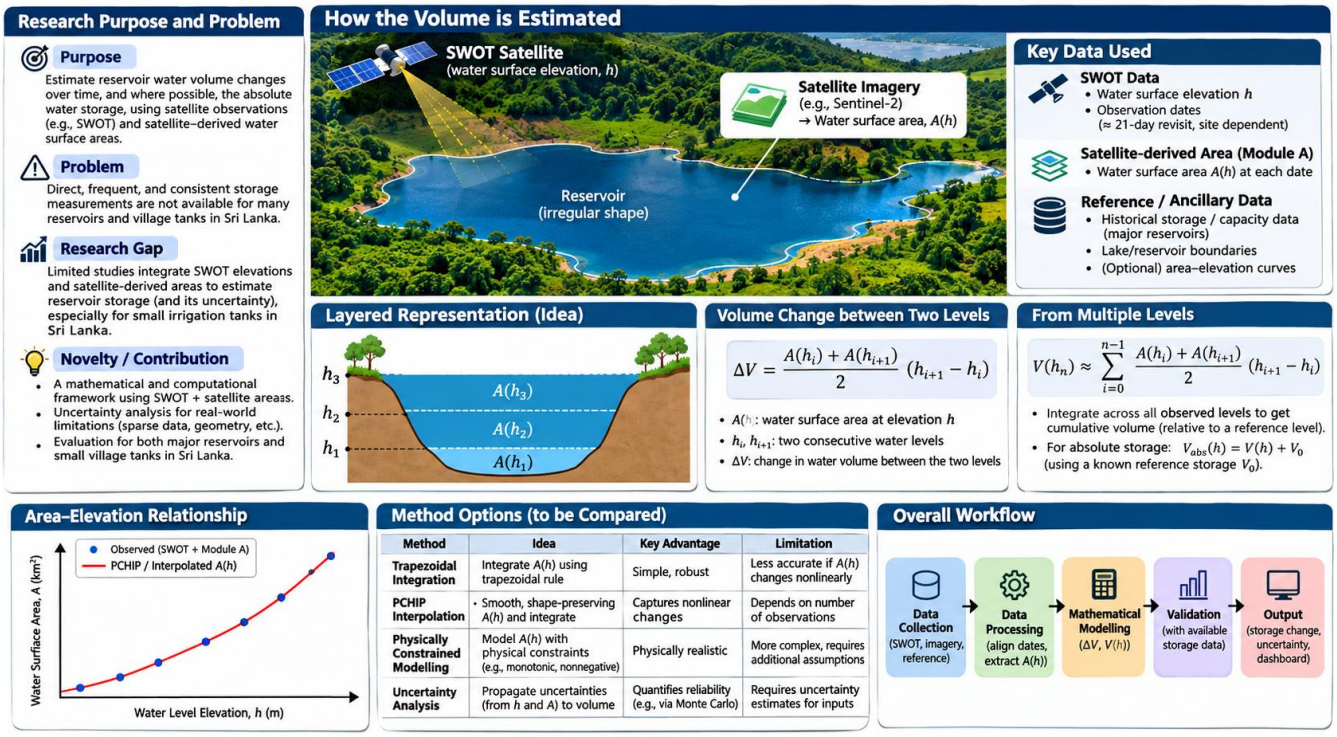



**Figure 1. Conceptual Module B overview — satellite heights, Module A areas, area–elevation integration, method comparisons and validation.**

**Core distinction** A surface elevation indicates the water level relative to its vertical reference. It does not directly measure the reservoir bottom elevation or total water depth.

# 5. Mathematical foundation and candidate methods

## 5.1 Fundamental storage relationship

Let h denote reservoir water-surface elevation in metres (on a consistent vertical datum), A(h) the horizontal inundated water area in $m^2$ at elevation h, and V(h) the stored water volume in $m^3$. For a connected, approximately level water body with fixed geometry:

$dV / dh = A(h)$ and $\Delta V(h_1,h_2) = \int_{h_1}^{h_2} A(z)\,dz$

The integration measures the water-filled horizontal cross-sectional area of thin layers. Rocks, protruding banks and irregular underwater shapes are naturally accounted for if A(h) is correctly known throughout the elevation interval; their 3D geometry does not need to be separately reconstructed. Endpoint areas alone, however, do not identify the exact intermediate curve.

## 5.2 Method A — trapezoidal integration (baseline)

For adjacent observed elevations $h_i$ and $h_{i+1}$ with areas $A_i$ and $A_{i+1}$:

$\Delta V_i \approx [(A_i + A_{i+1}) / 2] \times (h_{i+1} - h_i)$

This assumes linear change in area between the two heights. It is easy to reproduce and interpret, but can be biased when area changes sharply between observations. Example: $A_1$ = 80,000 $m^2$, $A_2$ = 100,000 $m^2$ and Δh = 1 m imply ΔV ≈ 90,000 $m^3$. This is an illustrative arithmetic example.

## 5.3 Method B — PCHIP interpolation and integration

Sort valid (h, A) pairs by increasing h, resolve duplicate/noisy levels, fit an area–elevation function using a piecewise cubic Hermite interpolating polynomial, then integrate that fitted curve over $[h_1,h_2]$. SciPy provides PchipInterpolator and its integrate(a,b) method. PCHIP preserves monotonicity when its input data are monotone, but raw noisy input can still be physically implausible. With only two unique points, interpolation reduces to a straight segment, so PCHIP is not intrinsically superior to trapezoidal integration. [4]

### 5.4 Proposed advanced method — constrained, uncertainty-aware A(h)

Fit a non-negative, non-decreasing area–elevation function for a fixed reservoir geometry: $A(h) \geq 0$ and $A(h_2) \geq A(h_1)$ whenever $h_2 > h_1$. Use measurement quality and uncertainty weights to reduce undue influence of unreliable pairs. Integrate the fitted curve and quantify uncertainty. Possible implementation: weighted monotone regression/isotonic fitting followed by PCHIP, or a constrained smooth optimization model (e.g., CVXPY), compared against direct PCHIP and trapezoids. The best approach must be decided experimentally, not presumed.

### 5.5 Storage change versus absolute storage

Net change between two levels does not require the exact bottom depth. Absolute storage needs a reference:

$V(h) = V(h_0) + \int_{[h_0 \to h]} A(z)\, dz$

Without a known $V(h_0)$, capacity relationship, bathymetry or a defensible zero-volume elevation, infinitely many absolute curves can share the same observed storage changes. The system will label absolute volumes as calibrated estimates or exploratory scenario ranges, never as directly observed totals.

### 5.6 Observation quality and vertical datum

SWOT heights, official gauge stage and capacity-curve elevations may use different vertical reference systems. A constant datum offset does not affect Δh itself, but it can corrupt cross-source pairing and height–storage curve evaluation. Confirm geoid/ellipsoid/benchmark conventions, time zones, units and reservoir identity. Prefer quality-screened height–area pairs rather than blindly averaging all available observations.

## 6. Technical questions, findings, proposed handling and limits

### 6.1 Unknown bottom elevation

Finding: SWOT measures the water surface, not underwater depth.

Proposed handling: Estimate changes by integrating A(h) between observed levels; anchor total storage with an external reference.

Remaining limitation: Total volume is not identifiable without reference information.

### 6.2 Large shoreline rock and unknown 3D shape

Finding: A rock displaces water, but each accurate horizontal water-covered area excludes occupied rock cross-section.

Proposed handling: Model area as a function of water elevation; test interpolation uncertainty across height gaps.

Remaining limitation: Two endpoint areas do not reveal the exact intervening rock/bank geometry.

### 6.3 Approximately horizontal surface

Finding: Reservoir surface is often treated as level at each observation.

Proposed handling: Use one representative water-surface elevation per connected reservoir, after quality checks.

Remaining limitation: Wind setup, flows, connectedness and satellite error create deviations.

### 6.4 Irregular banks and expanding area

Finding: Water area often increases with height but the rate is not constant.

Proposed handling: Compare trapezoids, PCHIP and physical constraints; stratify results by elevation interval.

Remaining limitation: Sharp unseen geometry transitions can still be missed.

### 6.5 Sparse and irregular satellite dates

Finding: SWOT has a nominal 21-day repeat cycle; local revisit opportunities vary.

Proposed handling: Use actual per-lake timestamps; distinguish temporal gaps from height gaps.

Remaining limitation: Unseen temporary floods or drawdowns cannot be reconstructed from endpoints alone.

### 6.6 Large elevation gaps

Finding: Interpolated A(h) over a broad interval may miss shoreline transitions.

Proposed handling: Gap-based stress tests, leave-out observations, uncertainty intervals and abstention flags.

Remaining limitation: A method cannot guarantee exact volume with insufficient geometrical information.

### 6.7 Noisy area or height measurements

Finding: Upstream Module A and SWOT measurement errors propagate to ΔV.

Proposed handling: Quality flags, plausible ranges, uncertainty propagation using Monte Carlo/bootstrap and sensitivity analysis.

Remaining limitation: Input error distributions must be justified, not guessed as certainties.

### 6.8 Storage definition mismatch

Finding: Official effective/live storage may differ from total physical water volume by a constant dead-storage offset.

Proposed handling: Confirm definitions; compare changes where offsets cancel and check whether geometry/time assumptions agree.

Remaining limitation: Sedimentation, datum changes or changing capacity tables may complicate comparison.

### 6.9 Small-tank data scarcity

Finding: Two tanks reportedly have reference information, but type and dates are unknown.

Proposed handling: Verify whether time-series storage or only design capacity is available; hold out truly independent data.

Remaining limitation: Capacity-only records cannot validate a time series of storage changes.

### 6.10 Uncertain generalization

Finding: One large reservoir's fitted A(h) must not be copied to another reservoir.

Proposed handling: Transfer modelling procedure; refit per reservoir; evaluate leave-one-reservoir-out performance where feasible.

Remaining limitation: Two small test tanks support limited claims of generalization.

### 6.11 Missing extreme events

Finding: A flood peak between passes may be unseen even when net endpoint change is estimated well.

Proposed handling: Separate endpoint net change from maximum storage and daily dynamics; mark missing-event risk.

Remaining limitation: No method can infer the unseen hydrograph uniquely without added observations/model assumptions.

## 7. Proposed IT system and technology choices

Pipeline: data acquisition → lake-ID harmonization and quality screening → date/time and datum alignment → cleaned h–A pairs → mathematical fit → volume-change integration → uncertainty propagation → independent validation → export/API/dashboard.

| Layer | Practical tooling | Why it is used |
|---|---|---|
| **Data acquisition** | Python, NASA Earthaccess, file APIs | Reproducible downloads and provenance. |
| **Data wrangling** | Pandas, NumPy, Parquet | Date joins, units, flags and numerical arrays. |
| **Geospatial integration** | GeoPandas, Shapely / Rasterio if needed | Reservoir polygons, masks and geographic matching. |
| **Numerical core** | SciPy PchipInterpolator; NumPy | Interpolation, numerical integration, simulations. |
| **Constrained modelling** | Scikit-learn isotonic regression or CVXPY | Physically plausible A(h) alternatives. |
| **Evaluation** | Scikit-learn metrics, SciPy, custom scripts | Reference comparison, uncertainty calibration and experiments. |

| | | |
|---|---|---|
| **Engineering** | Git/GitHub, Pytest, configuration files | Versioning, repeatability and quality control. |
| **Delivery (optional)** | FastAPI, Streamlit | Model-as-service and interactive plots/results. |

### 7.1 Does Module B need machine learning?

No. A strong IT final-year contribution can centre on scientific computing, reproducible data pipelines, numerical methods, constrained optimization and testable uncertainty quantification. Optional supervised ML (e.g., XGBoost residual correction or SWOT observation quality scoring) should be attempted only if reliable labels, adequate sample size and leakage-safe independent test data exist. Do not add ML solely to make the project appear more technical.

### 7.2 Recommended outputs

- For each lake and observation pair: $h_1$, $h_2$, $A_1$, $A_2$, method, estimated signed $\Delta V$ ($m^3$), uncertainty interval, observation-quality and gap flags.
- Where reference calibration is available: estimated absolute V(h), capacity percentage and uncertainty, clearly labelled calibrated rather than directly observed.
- For each experiment: matched sample count, MAE, RMSE, signed bias, relative error when denominator is meaningful, interval coverage, lake and date split.

## 8. Validation, comparisons and experimental design

6. First verify exact date/time alignment among SWOT observations, Module A outputs and independent official storage records; document tolerances and exclude invalid comparisons.
7. Construct independent reference $\Delta V = Vref(t_2) - Vref(t_1)$. Ensure the observed storage series uses a consistent definition, and report how close each reference time is to the satellite time.
8. Compare a constant-area approximation (diagnostic), trapezoidal baseline, direct PCHIP, constrained model and, if compatible, SWOT-provided storage-change estimates.
9. Evaluate at the same held-out dates and reservoirs. Do not tune model parameters on final test observations. Use chronological blocking and reservoir-level holdouts where available.
10. Report MAE, RMSE and bias of $\Delta V$; normalize errors by known reservoir capacity or characteristic storage scale where justified. Avoid percentage errors near zero actual changes.
11. Test area/height uncertainty sensitivity, elevation-gap bins, temporal-gap bins, small-sample conditions and coverage of declared uncertainty intervals. Clearly distinguish model uncertainty from missed-event uncertainty.
12. Transferability: fit the method with each small tank's own h–A samples and compare with genuinely independent tank records; do not transfer the shape curve from Mahakandarawa unchanged.

Feasibility gate: publication of some storage values establishes potential validation evidence, not a complete matched dataset. The 2024 table has five discrete dates; daily series and small-tank records still need retrieval, provenance checks, and assessment of paired observations.

## 9. Step-by-step implementation plan

| Phase | Work package | Decision / deliverable |
|---|---|---|
| **0 — Current** | Confirm lake IDs, data permissions, time ranges and teammate contributions. | Data inventory with status, quality and provenance. |
| **1 — Data feasibility** | Download SWOT files; inspect quality fields; obtain Module A areas and official records; match dates/datum. | Count of usable aligned observations, missingness audit and validation feasibility. |
| **2 — Baselines** | Implement trapezoid and direct PCHIP on observed height–area pairs. | Reproducible baseline results and plots. |
| **3 — Advanced method** | Test monotonic/physically constrained fit and uncertainty propagation. | Ablation experiments and uncertainty intervals. |
| **4 — Independent testing** | Major-reservoir accuracy tests; available small-tank holdouts; gap sensitivity. | Results table, failure analysis, honest confidence limits. |

| | | |
|---|---|---|
| **5 — Exploration** | Calibrate total storage where reference information permits. | Absolute-storage scenarios with documented assumptions. |
| **6 — Software output** | Package model, tests, API/dashboard as project time permits. | Runnable prototype, documentation and demonstration. |

## 10. Risks, safeguards and go/no-go decisions

| Risk | Practical safeguard |
|---|---|
| **Few same-date satellite/reference pairs** | Audit before modelling; widen documented time window or use multiple reservoirs; never fabricate validation. |
| **Area segmentation and water-under-vegetation uncertainty** | Request Module A confidence or error estimates and propagate rather than quietly correct segmentation. |
| **No small-tank storage time series** | Treat transfer as qualitative or limited consistency check; do not claim accuracy validation from capacity alone. |
| **Unknown reference datum or reservoir storage definition** | Discuss with engineer and data publisher before comparisons. |
| **No calibration for absolute storage** | Report changes only; offer explicit sensitivity scenarios, not definitive total volume. |
| **Complex method fails to outperform baseline** | Report negative result openly; it remains informative if experiments are well designed. |
| **Model overfitting / ML data shortage** | Reserve holdout periods/lakes; prefer simpler methods unless evidence supports complexity. |

## 11. Questions to confirm with engineer, supervisor and teammates

13. What exact quantity does each official storage series report: gross, effective/live or usable storage? Are its rating curves and reference datum available?
14. Do the two reported small tanks have dated storage readings, stage–storage curves, bathymetric surveys or only design/full-supply capacities? Which dates and units?
15. Can Module A deliver dated water area estimates for the same reservoir boundary and overpasses, plus area confidence/uncertainty?
16. Which SWOT LakeSP product version, lake identifier, vertical datum and quality flags are suitable for the selected reservoirs?
17. Is independent daily reservoir storage obtainable for 2023–2026 and what happens on days without new gauge readings?
18. What level of error is operationally meaningful ($m^3$, MCM or % of capacity), and how should uncertainty be communicated to irrigation managers?
19. For total storage, which reference value / capacity curve is scientifically acceptable and has the same definitions and datum?
20. Should optional ML be considered only after training-set size and benchmark performance have been established?

## 12. Expected outputs and contribution claims

Expected project outputs (not achieved results): (i) documented harmonized data workflow; (ii) baseline and advanced area–elevation integration models; (iii) independent reference comparisons for major reservoirs; (iv) transferability study for small tanks where ground truth permits; (v) uncertainty and sparse-observation analyses; and (vi) reproducible code, tests and optional visualization service.

Defensible intended contribution: evidence on the relative reliability and transferability of transparent storage-change estimation algorithms for Sri Lankan tanks when satellite elevations and satellite-derived areas are sparse or uncertain. Any claim that the proposed method is more accurate than existing methods must await comparative results.

## 13. References and source-verification notes

[1] Sri Lanka Irrigation Department. Administration Report 2024, Table 6-2, “Water availability in Major reservoirs”, p. 82. https://irrigation.gov.lk/web/images/Publications/Administration_Report_2024.pdf

[2] Imbulana, K. A. U. S. (2022). Depth-Area-Storage Capacity Relationships of Village Tanks. ENGINEER 55(2), 53–61. DOI: 10.4038/engineer.v55i2.7508. https://engineer.sljol.info/articles/7508/

[3] NASA/PO.DAAC SWOT Lake Single Pass (LakeSP) product documentation: lake water elevations, surface areas, quality fields and product-derived storage-change variables; verify product/version-specific descriptions before use. https://podaac.jpl.nasa.gov/SWOT

[4] SciPy documentation. scipy.interpolate.PchipInterpolator and integrate. https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.PchipInterpolator.html

[5] Team proposal / discussion notes (26 October 2026, not independently re-opened for this document): candidate lake IDs, usable SWOT observation counts and reported two-small-tank data lead. Source must be checked against the original proposal and actual files.

## Appendix A — Discussion-ready one-minute explanation

“Module B estimates the change in water stored in a reservoir from satellite-observed water level and water surface area. Rather than assuming a constant shape, we fit an area–elevation relationship and integrate it to estimate change in volume. We will compare a transparent trapezoidal baseline against PCHIP and a proposed uncertainty-aware physically constrained approach. We then test predictions against independent Sri Lankan reservoir records, investigate how sparse satellite observations and irregular geometry affect error, and evaluate whether the same modelling procedure works for small village tanks. Total volume is a secondary objective that requires a storage reference or additional bathymetric assumptions. The work is still at data-collection and method-design stage.”